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Modeling and Simulation

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M4 Documentation

Introduction

This project is a simulation of the electronics repair department, Manufacturing Support, that I work in. I am using SimPy to model a discrete event system that takes us through the four sources for jobs to funnel into the department. The customer RMAs, Advance Exchanges, Reships, and Production failures arrive based on my department's real historical arrival rate, and their outcomes (repaired, scrapped, return to vendor) are also calculated from historical close rates. The jobs wait for availability of technicians, technician's capability, and workstations in order to be evaluated and repaired. The workload is constrained to business hours of 7am-3:30pm, Monday-Friday. The preemption takes place for Advanced Exchanges, Reships, and Production failures that can interrupt in-progress repairs of customer RMAs but do not preempt each other. Technicians and stations with specialty capability can work on general jobs but specialty jobs are restricted to those specialty technicians and stations.

I am running these simulations in order to determine which parameters of the Manufacturing Support Department actually make an impact on performance. Throughout my testing, there were parameters that made a substantial difference over others. With this

information, I also want to be able to determine the performance of the department under different scenario configurations. These included higher demand, no preemption, base line, etc. The goal is determining the most efficient and effective combination of resources for the department.

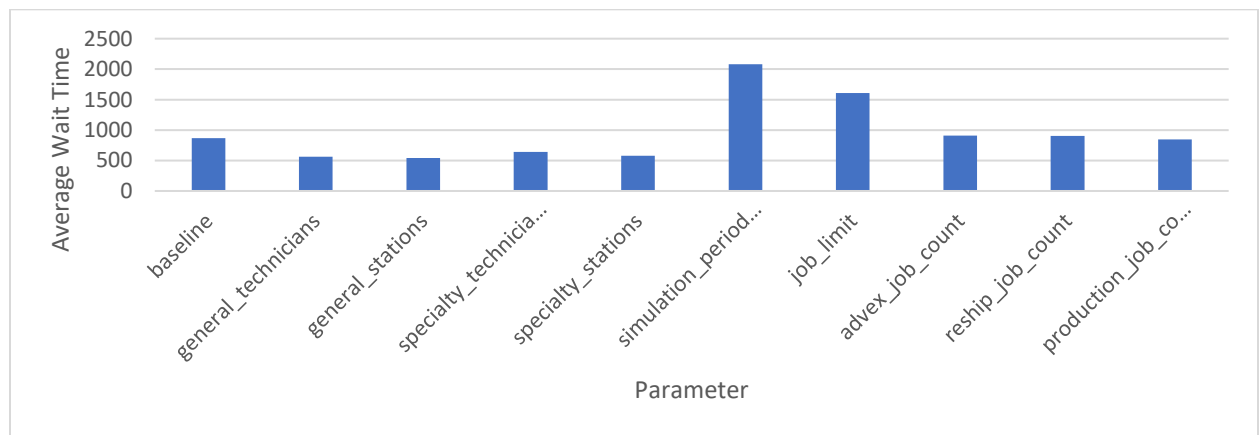
In my M3 preliminary results, I was able to determine that not every parameter makes a difference at baseline scenario configuration. In my M3 report, I found that with baseline demand, adding more resources had 0 impact on any metric. The results were exactly the same because resources were never the issue. In this milestone, I determined a flaw in my coding logic. In my real department, specialty technicians and stations are also able to evaluate general jobs but that was not correctly reflected in the logic. They were only being assigned specialty jobs. This meant that my M3 results had jobs sitting idle because they were not being assigned correctly when a specialty technician/station was available. This was corrected in my M4 deployment.

Parameter Analysis

I tested 9 parameter updates against baseline configuration of 1 general technician, 1 general station, 1 specialty technician, 1 specialty station, and 720hrs (30-day simulation of job flow). In my testing, I ran a 100% increase in each of the parameters. To further improve accurate results, each parameter update is run 100 times with the same random seeds, so each run encounters the same job rate and isolates the parameter change.

What I determined was that a single job source had the greatest impact on the department. Doubling advanced exchanges, reships, or production failures barely have an impact on wait time compared to baseline. Doubling customer RMAs have the single largest impact in my testing. The wait time is almost doubled just from changing this one parameter. This is because customer RMAs are the highest percentage of jobs flowing through the department in a 30-day simulated period while also taking the longest average time for evaluation compared to the other jobs. A shared increase in demand of jobs through the department is sustainable but an increase in customer RMAs puts a strain on the department's ability to evaluate and repair jobs in efficient timeline.

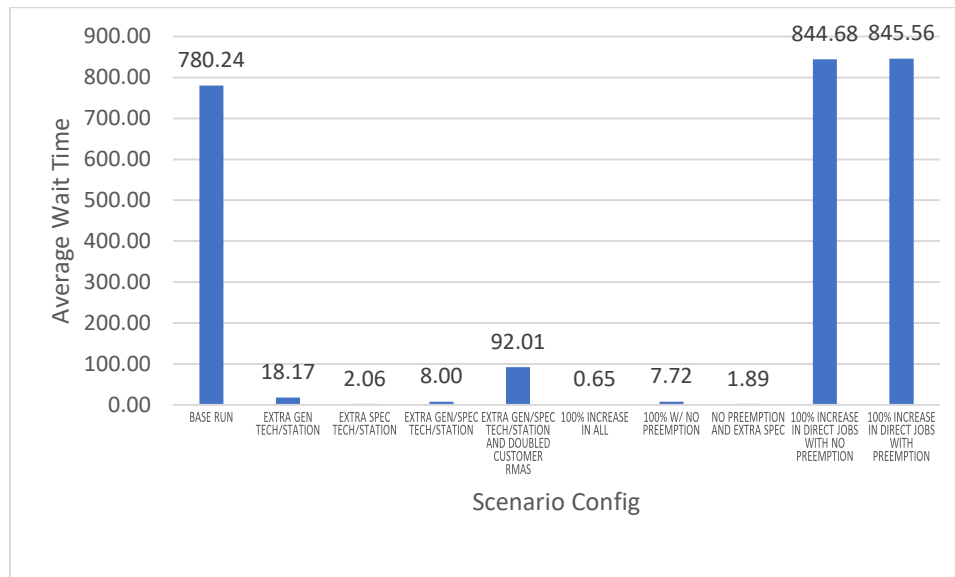
I: parameter_update.csv



Scenario Analysis

In scenario testing, I ran 10 different scenario configurations with multiple parameters changed to reflect real world situations.

II: Summary.csv



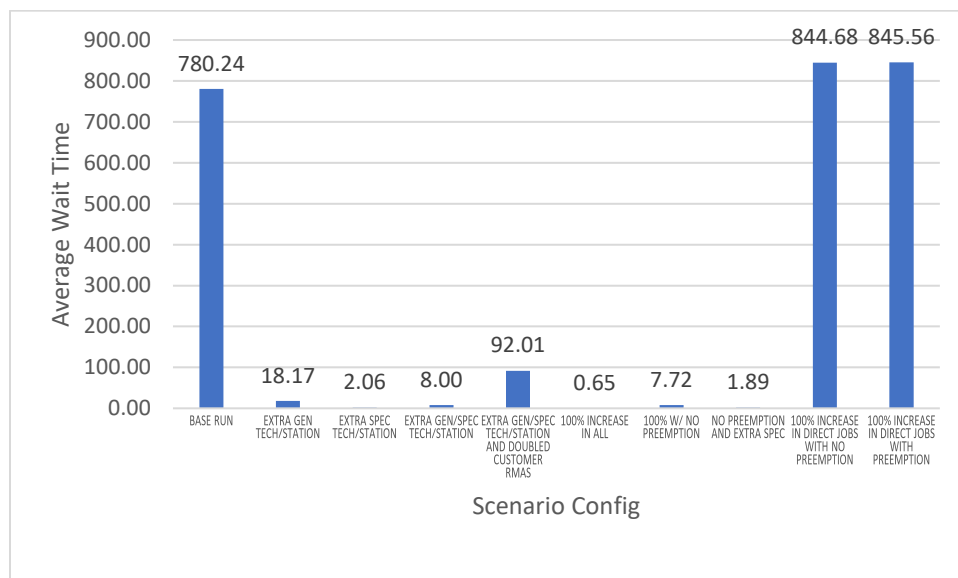
The immediate thing that caught my eye was the drastic drop in waiting time when resources were added compared to baseline. When configured for baseline of 1 general technician/station, 1 specialty capable technician/station, normal demand of 28 Customer RMA, 8 AdvEx, 8 reships, and 19 production failures within a 720hrs(30day) simulation time, wait times are 780hrs. When an extra general capable technician and station are added, waiting times drop ~97%. Waiting times drop by ~99% when an extra specialty capable technician and station are added instead. This is because specialty capable technicians/stations can also evaluate general jobs, but general capable technicians/stations cannot evaluate specialty jobs. The next thing to catch my eye was that even with resources doubled, when customer RMAs were doubled, waiting times exploded again. The increase in customer RMAs is the single biggest stress on the department. Doubling direct jobs, whether preemption was on or off, did not cause significant change in waiting times. This is because once the resources are running at capacity, jobs that take precedence stop mattering as there are no available resources to

continue evaluating. The parameter that improves waiting times is resource availability. Once the resources are at capacity, preemption does not improve waiting times.

Validation

With face validation, wait times and queues should grow and shrink with resource availability in my manufacturing simulation. We see that happen when comparing baseline configurations runs and when we add resources and/or increase demand. When general technicians/stations are added, we see a massive improvement in waiting times and queue length for resources. This is again seen with further improvement when specialty capable technicians/stations are added as they are able to perform both general and specialty jobs. We see the waiting times and queue grow when demand is increased and resources are no longer able to keep up with demand.

III: Summary.csv



IV: time.csv

scenario	sim_time	day_time	tech_queue_length	station_queue_length	techs_busy	techs_idle	stations_busy	stations_idle
base run	685	Day 28 - Monday 13:00	28		1	1	1	1
base run	690	Day 28 - Monday 18:00	28		1	1	1	1
base run	695	Day 28 - Monday 23:00	28		1	1	1	1
base run	700	Day 29 - Tuesday 04:00	28		1	1	1	1
base run	705	Day 29 - Tuesday 09:00	30		1	1	1	1
base run	710	Day 29 - Tuesday 14:00	31		1	1	1	1
base run	715	Day 29 - Tuesday 19:00	31		1	1	1	1
base run	720	Day 30 - Wednesday 00:00	31		1	1	1	1

V: time.csv

scenario	sim_time	day_time	tech_queue_length	station_queue_length	techs_busy	techs_idle	stations_busy	stations_idle
Extra Gen Tech/Station	685	Day 28 - Monday 13:00	0	0	0	1	2	1
Extra Gen Tech/Station	690	Day 28 - Monday 18:00	0	0	0	1	2	1
Extra Gen Tech/Station	695	Day 28 - Monday 23:00	0	0	0	1	2	1
Extra Gen Tech/Station	700	Day 29 - Tuesday 04:00	0	0	0	1	2	1
Extra Gen Tech/Station	705	Day 29 - Tuesday 09:00	0	0	0	2	1	2
Extra Gen Tech/Station	710	Day 29 - Tuesday 14:00	0	0	0	2	1	2
Extra Gen Tech/Station	715	Day 29 - Tuesday 19:00	0	0	0	2	1	2
Extra Gen Tech/Station	720	Day 30 - Wednesday 00:00	0	0	0	2	1	2

VI: time.csv

scenario	sim_time	day_time	tech_queue_length	station_queue_length	techs_busy	techs_idle	stations_busy	stations_idle
100% increase in direct jobs with preemption	685	Day 28 - Monday 13:00	30		1	1	1	1
100% increase in direct jobs with preemption	690	Day 28 - Monday 18:00	31		1	1	1	1
100% increase in direct jobs with preemption	695	Day 28 - Monday 23:00	31		1	1	1	1
100% increase in direct jobs with preemption	700	Day 29 - Tuesday 04:00	31		1	1	1	1
100% increase in direct jobs with preemption	705	Day 29 - Tuesday 09:00	34		1	1	1	1
100% increase in direct jobs with preemption	710	Day 29 - Tuesday 14:00	35		1	1	1	1
100% increase in direct jobs with preemption	715	Day 29 - Tuesday 19:00	35		1	1	1	1
100% increase in direct jobs with preemption	720	Day 30 - Wednesday 00:00	35		1	1	1	1

Parameter validation is done through using my department's real-world data. The data is cleaned up to hide sensitive data, but the arrival rates, service times, and outcomes are pulled from real data. I used my department's data to find the average of how often jobs arrive, how long repairs take, which jobs are general or specialty, outcome, and priority. The only exception being reshipe jobs. These are not logged but from my personal experience working in the department, they mimic the same data as advanced exchanges, so they implement the same rates in the logic.

Statistical Summary

Replication_summary.csv

metric	mean	std_dev	min	max	ci_lower	ci_upper
Average Wait Time (hrs)	982.6487	1459.8927	0.7438	8548.5406	696.5097	1268.7877
Average Turnaround Time (hrs)	1080.3254	1540.8208	8.7656	8949.8235	778.3245	1382.3262
Throughput (jobs/hr)	0.0251	0.0182	0.0026	0.0606	0.0216	0.0287

For these metrics, I ran the baseline configuration 100 times with different seeds to produce different outcomes. I then calculated the metrics across those runs. The difference between these metrics is massive. The wait times ranged from just under an hour up to ~8500hrs. All metrics show extreme variability between the minimum and maximum result. The confidence interval tightens these results to a more manageable window. The reason for the large gaps in results is due to the number of jobs, resources and the business hours constraint. A job received at the end of the work week may have to wait over the weekend and still be preempted by other jobs. This swings the results to both boundaries.

Conclusions

Resources are the bottleneck in my simulation/department. Once more general technicians/stations were added, waiting times dropped by 97% and waiting times dropped by 99% when specialty technicians/stations were added instead. Specialty technicians/stations were able to drop wait times more because they are able to complete general and specialty jobs. Preemption did not make a difference once the resources were at capacity. Customer

RMA demand is the parameter that inflicts the most stress on the department. When the department resources are doubled and demand at baseline, waiting time is around 8hrs, within a workday. When the department resources are still doubled and customer RMA demand doubled, waiting time is around 92hrs, a roughly 11x increase. Doubling customer RMA demand eliminates the benefit of having more resources. Doubling the direct jobs with no more resources did not make a significant difference to waiting times. This shows that there is space for direct jobs with baseline resources. The best scenario I tested was adding an extra specialty technician/station at normal demand. Wait times are ~2hrs and turnaround falls with ~37hrs. The worst scenario is normal resources with higher demand of customer RMAs. The biggest weakness I see with the model is the lack of accurate reship data from my real-world department. I believe the model would be more accurate with hard-data backing the average rates and outcomes. The model also fails to take into account normal human rhythm of working pace. The model simulates picking up a job immediately once one is completed for all 8 working hours of the day. The biggest strength of the model is that it is backed by real-world data and averages for the most common job types. If the department needed to change one thing, I would focus on adding more specialty technicians/stations as it was the greatest contributor to lowering waiting times and turnaround times.